

to see spectrum as something other than a physical resource to be licensed. Portions of the radio spectrum could instead be treated as a commons. A commons, like the air we breathe and the language we speak, is a shared, renewable resource. It is open to all. It is not completely free or inexhaustible, but it can seem that way if individuals follow rules to prevent over-grazing. A commons is entirely compatible with competitive capitalism. The key is that the marketplace occurs among users of the commons; the commons itself cannot be bought or sold. We have no trouble accepting the automobile and trucking industries, even though they depend on public roads and highways that are free to use and maintained by the government. And we accept that even though anyone can drive on the highway, everyone has to observe speed limits, seatbelt laws, and other safety rules. Those public roads even coexist with private toll roads, but we don't think that privatizing all the roads would improve the quality of transportation in the US.

A spectrum commons works just like the highways. Government defines the scope of the common resource and sets limited rules to facilitate efficient use. That means setting aside unlicensed frequencies, adopting rules to facilitate new “underlay commons,” setting power limits or other technical standards, and responding to any breakdowns.

The beauty of a spectrum commons is that it creates the right incentives. Exclusive licensing and propertization create spectrum monopolies, which seek to maximize the rents they can collect. Forcing licensees to buy spectrum at auction ensures it goes to those who value it highly, but it forces the winner to recoup its up-front investment, biasing the way it makes use of the spectrum. As noted above, exclusive licensing also encourages receiver manufacturers to make their devices as dumb as possible, while a spectrum commons has the opposite effect. In a commons environment, companies can respond to marketplace demands by tailoring new services, since the costs of entry are minimal.